

Practice Quiz: Processes, Threads, and Communication

(Based on Distributed Systems, 4th Ed. v01, Ch 03, Ch 04)

Part I: Multiple Choice

1. (2 points) Which best describes a thread?
 - A. A minimal software processor in whose context a series of instructions can be executed.
 - B. A hardware device for memory access.
 - C. The same as a process.
 - D. A hardware network interface.
2. (2 points) What is saved during a thread context switch?
 - A. Only the program counter and stack pointer.
 - B. The full processor context and any state necessary to resume execution later.
 - C. Network routing information.
 - D. Only the operating system kernel data.
3. (2 points) What is a key performance advantage of thread context switching versus process context switching?
 - A. Thread switches require a full memory remap and are slower.
 - B. Thread switches are generally faster since threads share the same address space within a process.
 - C. Process context switches use less CPU.
 - D. Both are the same speed.
4. (2 points) In the OSI network model, which layer is responsible for error and flow control at the frame level?
 - A. Physical Layer
 - B. Data Link Layer
 - C. Network Layer
 - D. Transport Layer
5. (2 points) Which of the following is *not* a typical purpose of middleware in distributed systems?
 - A. Naming protocols
 - B. (Un)marshaling of data

- C. Managing processor power states
 - D. Security protocols
6. (2 points) What does a client typically do in a transient synchronous client/server communication model?
- A. Continues processing after sending a request.
 - B. Waits for a reply after issuing a request.
 - C. Never communicates directly with the server.
 - D. Handles requests from the server.

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Part III: Matching (Write the letter of the correct description next to each term.)

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|-----------------------------|---|
| 1. Thread context | [label=.] |
| 2. Process | 1. Stores protocol for error and flow control at the frame level. |
| 3. Data Link Layer | 2. Stores the state required to resume a thread's execution. |
| 4. Middleware | 3. Provides shared services like naming, security, and data marshaling. |
| 5. Persistent communication | 4. A software context in which one or more threads execute. |
| | 5. Messages are retained until successful delivery. |

Answer Key

Part I: Multiple Choice:

1. A
2. B
3. B
4. B
5. C
6. B

Part II: Short Answer:

1. A thread is a minimal software processor for executing instructions; a process is a context that can hold one or more threads, providing isolated resources.
2. Persistent communication stores messages at intermediate servers until they are delivered, unlike transient communication in which messages are discarded if the receiver or next server is unavailable.
3. The transport layer provides the core data transmission facilities in distributed systems, handling reliable (TCP) or unreliable (UDP) transport of data between network hosts.
4. Examples: Naming protocols, (un)marshaling of data, security protocols.

Part III: Matching:	1. Thread context	B
	2. Process	D
	3. Data Link Layer	A
	4. Middleware	C
	5. Persistent communication	E